

Program 10: Demonstrate kubernetes Dashboard and CLI for cluster monitoring

Project Structure:

Pgm10

->deployment.yaml

->service.yaml

Step 1: Create deployment.yaml

```
GNU nano 7.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 2: Create service.yaml

```

GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort

```

Step 3: Apply deployment & service

```

1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created

```

```

1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg10$ kubectl apply -f service.yaml
service/nginx-service created

```

Step 4: Verify resource

```

1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg10$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2           51m

```

```

1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg10$ kubectl describe svc nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
Name:                nginx-service
Namespace:            default
Labels:               <none>
Annotations:          <none>
Selector:             app=nginx
Type:                 NodePort
IP Family Policy:     SingleStack
IP Families:          IPv4
IP:                   10.103.102.238
IPs:                  10.103.102.238
Port:                 <unset> 80/TCP
TargetPort:           80/TCP
NodePort:             <unset> 31065/TCP
Endpoints:            10.244.0.70:80,10.244.0.71:80
Session Affinity:     None
External Traffic Policy: Cluster
Events:               <none>

```

Step 5: Access the service

```

1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg10$ minikube service nginx-service

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31065

 Opening service default/nginx-service in default browser...

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Step 6 :Open Kubernetes Dashboard

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg10$ minikube dashboard
```

```
🐼 Verifying dashboard health ...
```

```
🚀 Launching proxy ...
```

```
🐼 Verifying proxy health ...
```

```
🐼 Opening http://127.0.0.1:43167/api/v1/namespaces/kubernetes-dashboard/services/http:kubern  
etes-dashboard:/proxy/ in your default browser...
```

```
Opening in existing browser session.
```

